



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO BALAKRISHNA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

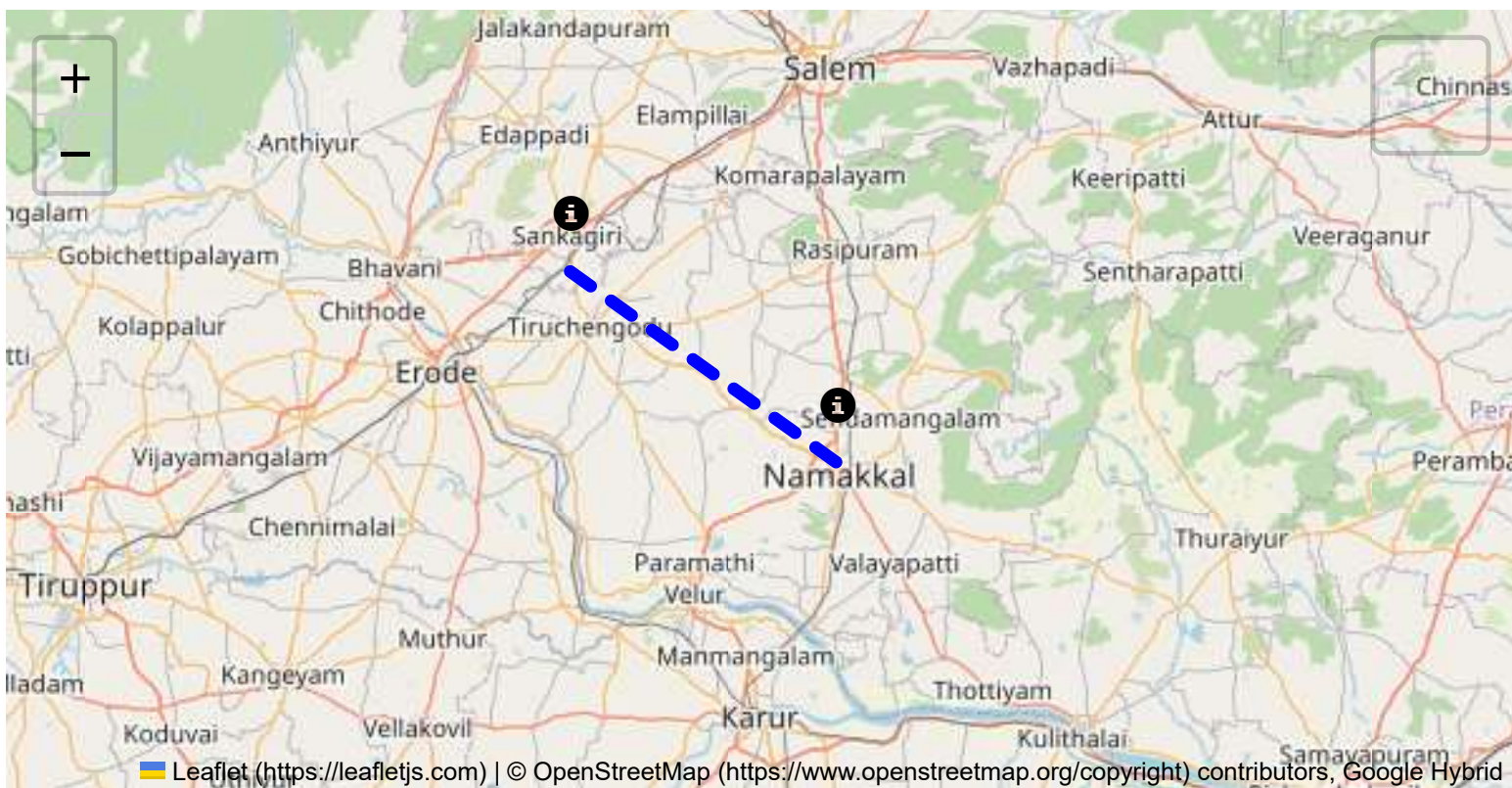
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

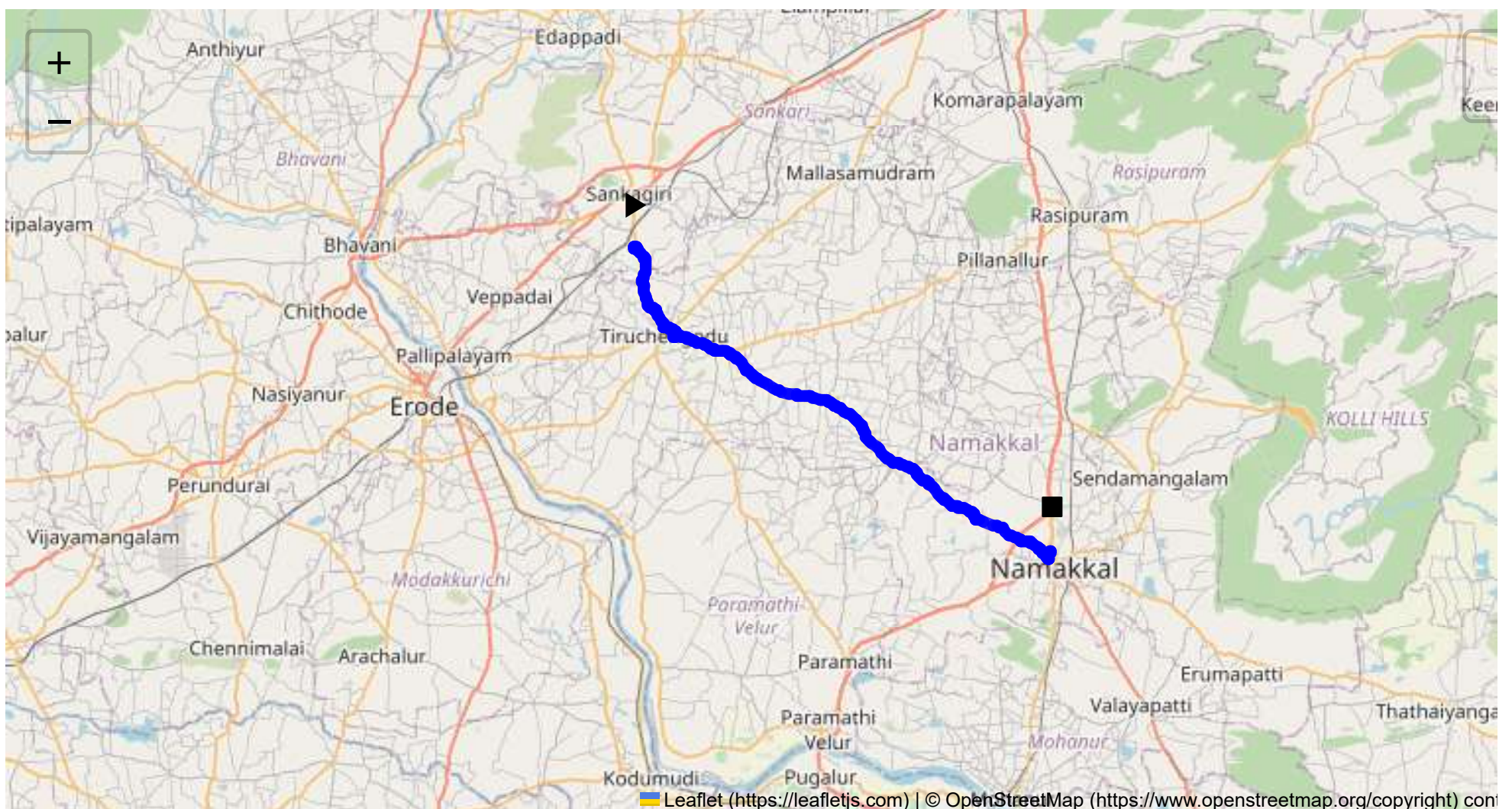
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.13 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.23141, 78.164757)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W in Sangagiri to R.P Pudur in Namakkal via Salem Road spans approximately 44.13 kilometers. It typically takes about 0.99 hours for heavy vehicles, including those carrying hazardous materials. The route predominantly follows NH44, which is a major highway known for connecting significant urban and industrial regions.

2. Typical Weather Conditions and Potential Hazards

The region experiences a tropical climate with high temperatures, especially between March and June. During the monsoon season (June to September), heavy rains can lead to slippery roads and poor visibility. Fog can occasionally affect driving conditions in the early mornings during winter months.

3. Traffic Patterns

Traffic tends to peak during morning hours (7 AM to 10 AM) and evening hours (5 PM to 8 PM) owing to regular commuter traffic. NH44 can experience congestion, particularly around interchanges and near urban centers like Salem and Namakkal.

4. Assessment of Road Quality and Infrastructure

NH44 is considered a well-maintained highway with good road quality, offering an adequate number of lanes to facilitate smooth traffic flow. However, periodic maintenance and construction work could lead to temporary delays.

5. Suggestions for Alternative Routes

In the event of an emergency or unforeseen congestion on NH44, an alternative route through SH86 that runs parallel for parts of the journey could be considered. This road might be less direct but can serve as a viable backup.

6. Summary of Local Regulations on Hazardous Material Transport

Tamil Nadu imposes strict regulations on the transport of hazardous materials, especially concerning vehicle compliance and timing restrictions. Transportation during night hours is often preferred to avoid peak traffic and reduce the risk of accidents in populated areas.

7. Overview of Historical Incidents

There have been reports of accidents involving heavy vehicles on NH44, often attributed to driver fatigue and speeding. These are not frequent but remain a point of concern for vehicles carrying hazardous materials.

8. Environmental Considerations and Sensitive Areas

The proposed route does not pass through any protected environmental zones, but care should be taken to prevent spills or emissions near agricultural lands, which are prevalent along the route.

9. Communication Coverage

Mobile network coverage along NH44 is generally reliable, although there may be occasional dead zones, particularly in stretches between major towns. It is advisable for drivers to have contact information for emergency services readily available and to keep radios operational for communication.

10. Estimated Emergency Response Times

Emergency response times can vary based on location along the route. Close to urban centers like Namakkal and Salem, response times could be as short as 15 to 30 minutes. In rural sections, it may take up to an hour, depending on road traffic and availability of services.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Namakkal poses moderate risks typical of major highways. The main concerns are weather conditions during monsoon, potential congestion during peak traffic hours, and regulatory compliance for vehicles transporting hazardous materials. Ensuring adequate preparation and safety measures can significantly mitigate these risks. Regular communication and contingency planning are essential, especially for trucks carrying hazardous materials. Overall, the route is considered safe, provided standard precautions are followed.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.22546, 78.16416	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
16	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
17	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
18	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
24	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
27	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
28	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
29	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
33	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
25	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
26	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
30	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.22546, 78.16416

